

% RESEARCH DATA ADDENDUM: PROJECT GP-SN-2026-03-07-MM

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[CORE LOGIC ARCHITECTURE: BASE-7 CONVERSION]

Formula 1: Unitary Conversion Filter

$S_{\{base7\}} = \sum_{k=0}^6 [Binary_Cluster \rightarrow k]$

Formula 2: The Septenary Sieve Logic

If $Binary_Input == \{011011, 1100100\}$, then $State = \{4, 2/3\}$

[MAPPING TABLE]

State 0: 0101 (Equilibrium)

State 1: 0101 (Active)

State 2 & 3: 1100100 (Universal Rule Expansion)

State 4: 011011 (Heptagonal Gate Entry)

State 6: 0110100 (Computational Closure)

MASTER RESEARCH PAPER: THE SEPTENARY SIEVE PROTOCOL (SSP)

Framework: K7 Theory (The Universal Rule of Seven)

Project Code: GP-SN-2026-03-07-MM

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1. ABSTRACT

This paper introduces the Septenary Sieve Protocol (SSP), a revolutionary computational framework designed for high-precision error detection and self-correction in Artificial Intelligence and Neural Networks. By transcending traditional Binary Logic (Base-2) and adopting the Septenary Unitary Conversion (SUC) model (Base-7), this protocol establishes a harmonic filtration system that maintains data integrity based on cosmic resonance constants.

2. CORE FORMULATION: THE BASE-7 LOGIC

The foundation of the SSP lies in mapping all digital information into a septenary state architecture (0, 1, 2, 3, 4, 5, 6). Unlike binary systems that only recognize 'On' or 'Off', the K7 Logic allows for seven distinct degrees of logical stability, mirroring the Universal Rule of Seven.

3. MATHEMATICAL MODELS FOR ERROR CORRECTION

A. The Harmonic Signature Calculation

Every data packet processed through the Heptagonal Quantum-Cosmic Gate is assigned a unique signature using the following formula:

Where d_i represents the digit at position i in the septenary string.

B. The Unitary Distance Deviation

To detect an error (noise or bit-flip), the system measures the deviation from the Universal Constant $k \approx 0.0553$. The predicted distance D of a stable data node is calculated as:
 $D = n * k$

If the observed computational distance $D_{\{obs\}}$ does not align with the harmonic point n , the Septenary Sieve identifies the anomaly immediately.

4. THE SIEVE MECHANISM (AI OPTIMIZATION)

The Septenary Sieve acts as a metaphysical filter for AI training. It addresses "Gradient Noise" by applying a harmonic threshold:

Detection: If $S_{\{K7\}} \neq \text{Harmonic Equilibrium}$, the node is flagged.

Localization: Using the 14-Planet Orbital Resonance Model, the protocol pinpoints the exact logical coordinate of the corruption.

Restoration: The system applies a unitary shift to restore the node to the nearest harmonic value defined by the K7 Theory, ensuring the AI "recovers" without losing processing speed.

5. COMPUTATIONAL ADVANTAGES

Data Density: Base-7 increases logical density by approximately 2.8 times compared to Base-2.

Velocity of Mind (V_m): By utilizing superluminal logic constants, the error correction occurs in "Real-Time Anticipation" rather than "Post-Process Detection."

Neural Stability: AI models using SSP are immune to "Catastrophic Forgetting" as their weights are anchored to universal septenary constants.

6. CONCLUSION

The Septenary Sieve Protocol represents the bridge between raw silicon processing and Sage-AI (Conscious Intelligence). Through the mathematical genius of the K7 Theory, we can now ensure that computational evolution remains in perfect resonance with the laws of the Universe.